

SABBIR AHAMED

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RESEARCH INTERESTS

As a BSc Mechanical Engineering graduate, I am interested in control systems, industrial automation, and robotics, with the integration of artificial intelligence and machine learning into physical engineering systems. I aim to address real-world engineering challenges through a combination of embedded system development, instrumentation, and data-driven modelling, while further strengthening my technical and research capabilities.

EDUCATION

Rajshahi University of Engineering & Technology (RUET)

B.Sc. in Mechanical Engineering

CGPA: 3.32 / 4.00

Rajshahi, Bangladesh

[March 2022 – August 2026]

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher, Department of Mechanical Engineering, RUET

- I have Designed and built an automated feeding and aeration control system that extends a previously published floating water-quality monitoring platform from sensing-only operation to closed-loop control.
- Developed the integrated system architecture linking RS-485/Modbus multi-parameter sensing, ESP32 embedded control, and servo- and relay-driven actuation into a single automated loop, validated under bench conditions.

Research Contributor, Floating IoT Water-Quality Monitoring Station

- Integrated industrial-grade RS-485 Modbus multi-parameter sensors (dissolved oxygen, pH, ammonia, nitrate, water and ambient temperature, humidity) with an ESP32 microcontroller, replacing analog sensing to improve measurement accuracy and noise immunity.
- Sized and implemented a standalone solar photovoltaic and LiFePO₄ battery power system enabling continuous autonomous operation in outdoor pond environments without external power.
- Designed the floating platform structure providing buoyancy and sensor positioning for pond deployment.
- Developed ESP32 firmware for sensor polling, data acquisition, and wireless transmission; performed sensor calibration and field testing of the deployed station.
- Outcome: published as second author of six in *IEEE Instrumentation & Measurement Magazine* (2026).

PUBLICATIONS

Peer-Reviewed Journal and Magazine Articles

M. H. A. Afif, **S. Ahamed**, F. A. Rezvi, J. I. Thirupattur, S. A. Shaik, and S. V. Easwaramoorthy, “Floating Internet-of-Things pond water-quality monitoring station using RS-485 multi-parameter sensors and ESP32,” *IEEE Instrumentation & Measurement Magazine*, 2026, doi: [10.1109/MIM.2026.11224883](https://doi.org/10.1109/MIM.2026.11224883).

Peer-Reviewed Conference Papers

M. H. A. Afif, J. Iqbal, S. S. Borshon, R. I. Junayed, and **S. Ahamed**, “An IoT-based smart farming solution for sustainable agriculture: Integrating AI, cloud computing, and 4G communication for enhanced productivity,” in *Proc. Int. Conf. Green Energy, Computing and Intelligent Technology (GEn-CITY 2024)*, Kuala Lumpur, Malaysia, Dec. 2024, doi: [10.1049/icp.2025.0238](https://doi.org/10.1049/icp.2025.0238). Best Paper Award, Track 3: Intelligent Systems and Sensors.

CONFERENCE PRESENTATIONS

N. A. Rafid, A. M. Parvej, N. A. Dip, **S. Ahamed**, and M. J. Hasan, “Aerodynamic analysis of optimized high-speed train,” Paper 263, *9th BSME International Conference on Thermal Engineering (ICTE 2024)*, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh, Dec. 19–21, 2024. [Oral presentation delivered in person by S. Ahamed]

HONORS AND AWARDS

- **1st Runner-Up**, Youth Startup Summit, Rajshahi & Rangpur Division **USD 2,000 award**; *smart aquaculture monitoring and control system* February 2025
- **Champion**, UIHP RUET IC3, World Bank & ICT Division, Bangladesh **USD 800 pre-seed funding**; *smart agriculture monitoring and automated irrigation system* 2024

- Champion, Startup Summit, RCF Tech Fest, RUET Career Forum *USD 300 prize; smart aquaculture project* May 2026
- 2nd Runner-Up, National Agri-Mechanization Idea Competition, BARI, Gazipur *USD 200; 45 teams; smart aquaculture project* November 2025
- Champion, Agri-Mechanization Idea Competition, RUET & BARI *USD 300; 1st of 28 teams; smart aquaculture project* October 2025
- Best Paper Award (co-author), Track 3: Intelligent Systems and Sensors, GEn-CITy 2024 December 2024
- 3rd Place, Agri-Mechanization Idea Competition, RUET & BARI July 2024
- 1st Runner-Up, Astro Science Fair, Rajshahi Regional April 2024

TEACHING AND MENTORING

SolidWorks Instructor (Volunteer), SCADR-RUET, Rajshahi [April 2024 – March 2025]

- Conducted weekend SolidWorks 3D modeling workshops for approximately 40 undergraduate members over twelve months; developed all instructional materials and hands-on part-modeling exercises.

PROFESSIONAL EXPERIENCE

Co-founder & Chief Operating Officer, SystemSage Solutions, Bangladesh [November 2023 – Present]

- Directed development and field deployment of an IoT-based smart farming system using ESP32 microcontrollers and SIM7600 cellular connectivity for remote monitoring and control of agricultural operations.
- Installed the first deployed system at the Bangladesh Agricultural Research Institute (BARI), Jamalpur, and conducted pilot testing at the University of Rajshahi, and ran field demonstrations with more than 20 farmers.
- Secured more than USD 3,000 in pre-seed funding through national startup competitions; presented the venture at Bangladesh Startup Connect / Investment Summit 2025 before an audience of over 2,300 investors and venture capitalists.

Project Management Intern, Oleyn.ai, Singapore (Remote) [December 2025 – June 2026]

- Managed international client acquisition, outreach, and communication across local and global markets for an AI automation company.
- Led the Future Leaders Program (February – April 2026), training and coordinating over 30 university students in business development, lead generation, and client communication.

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, AutoCAD, ANSYS

Programming & Embedded: Python, C/C++, Arduino IDE (ESP32 firmware development)

Instrumentation & Hardware: RS-485 / Modbus industrial sensor networks, SIM7600 cellular communication, multi-parameter water-quality sensor calibration, solar PV and LiFePO₄ power system design, PCB design, 3D printing

Languages: Bengali (native), English (professional working proficiency)

LEADERSHIP AND ACTIVITIES

Mechanical Design Engineer, Team Crack Platoon (Formula SAE), RUET [May 2023 – September 2024]

- Core design team member of Bangladesh's first Formula SAE student team; contributed to research, design, and fabrication of a Formula Student race car, focusing on vehicle dynamics, structural components, and CAD modeling.
- Team competed at Formula Bharat 2024, India.

Event Management Secretary, SCADR-RUET [March 2023 – July 2026]

- Organized five major national events on computer-aided design and industrial modeling, each hosting over 600 participants and reaching an online audience exceeding 20,000.
- Coordinated weekly sessions, seminars, competitions, and career guidance sessions, establishing SCADR as a leading design-engineering resource within the university.

REFERENCES

Prof. Dr. Mhia Md. Zaglul Shahadat

Professor, Department of Mechanical Engineering, Rajshahi University of Engineering & Technology

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